Figure 10.1 How the two planning prompt engineering strategies align with the various techniques

10.1 Understanding direct solution prompting

Direct solution prompting is generally the first form of prompt engineering that users employ when asking LLMs questions or solving a particular problem. Given any LLM use, these techniques may seem apparent, but they are worth reviewing to establish the foundation of thought and planning. In the next section, we'll start from the beginning, asking questions and expecting answers.

10.1.1 Question-and-answer prompting

For the exercises in this chapter, we'll employ prompt flow to build and evaluate the various techniques. (We already extensively covered this tool in chapter 9, so refer to that chapter if you need a review.) Prompt flow is an excellent tool for understanding how these techniques work and exploring the flow of the planning and reasoning process.

Open Visual Studio Code (VS Code) to the `chapter 10` source folder. Create a new virtual environment for the folder, and install the `requirements.txt` file. If you need help setting up a chapter's Python environment, refer to appendix B.

We'll look at the first flow in the `prompt_flow/question-answering-prompting` folder. Open the `flow.dag.yaml` file in the visual editor, as shown in figure 10.2. On the right side, you'll see the flow of components. At the top is the `question_answer` LLM prompt, followed by two `Embedding` components and a final LLM prompt to do the evaluation called `evaluate`.